

FEATURE ENGINEERING

Last week, we augmented our dataset by engineering multiple features. In the price dataset, we created stock price direction lagged variables and a stock price direction target variable. We plan to use the lagged variables to predict stock price direction with this data. In our sentiment dataset, we created the same stock price direction target variable. For this dataset, we will use news sentiment data to predict stock price direction. Finally, our combined dataset which contains both historical stock price and news sentiment polarity was augmented with lagged variables for both sentiment polarity and stock price direction. The same stock price direction target variable was also added to this dataset. We will use sentiment polarity as well as all of the lagged variables to predict stock price direction in this combined dataset. After the creation of these features, we did not see a need to add any more.

DIMENSIONALITY REDUCTION

This week, we considered reducing the dimensions of our datasets. However, the dimensionality of each is fairly small. For example, the price dataset only contains stock price direction and three lagged variables. Even still, we decided to perform principal component analysis (PCA) on the combined dataset. PCA allows us to transform many variables into a smaller set of uncorrelated variables that capture the maximum possible variance in the dataset. For example, we specified our PCA example to capture 99% of the variance in the combined dataset. While PCA is helpful in larger datasets with high dimensionality, it does not make as much sense to use in our case. Our data has very few features and our exploratory data analysis did not indicate we had significant problems with things like multicollinearity, which PCA can help mitigate. We will save the combined dataset with PCA performed as a later option for a feature set, but it is unlikely that we will use it in our models.